

AWS Lambda Support Ending for Python 3.9 – Migrate to Python 3.12

Why Python 3.12 is a good target [↗](#)

- **Performance:** Faster CPython implementation (significant improvements in startup and runtime performance).
 - **Longer support window:** Released October 2023, with support into 2028+.
 - **New syntax and features:** Improvements in error messages, f-string enhancements, isolated subinterpreters, etc.
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Considerations for Lambda Functions [↗](#)

1. AWS Lambda Support for Python 3.12

- Confirm [Lambda runtime availability](#). As of early 2025, Python 3.12 is supported.
- Update your function runtime in the console, CLI, or IaC (e.g., CloudFormation/CDK).

2. Lambda Layers Compatibility

- Any Lambda layer used **must be rebuilt** with Python 3.12 binaries.
- Common issue: layers built for 3.8–3.10 may not work due to binary incompatibility.
- Repackage or rebuild dependencies and native libraries with Python 3.12.

3. Third-Party Library Compatibility

- Double-check all libraries in your requirements (e.g., `requirements.txt`) for 3.12 support.
- Use virtual environments with Python 3.12 to pre-test package installs.

4. Container-based Lambdas

- If you're using custom containers, update your Dockerfile to base from a `python:3.12` image.
- Ensure libraries are compiled within that environment.

5. Testing and Deployment

- Unit + integration test after migration.
 - Consider rolling out changes using canary or linear deployment strategies via Lambda aliases.
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Suggested Migration Plan [↗](#)

1. Audit current Lambda functions:

- Identify runtime versions and layers used.
- Flag those using 3.8 or 3.9 (end of support approaching).

2. Build new Lambda layers:

- Rebuild for Python 3.12 using a compatible Docker or EC2 environment.
- Store in new ARNs for clarity.

3. Update CI/CD pipelines:

- Ensure Python 3.12 is installed in build environments (GitHub Actions, CodeBuild, etc.).
- Update `buildspec.yml` or equivalent files.

4. Gradual rollout:

- Move non-critical functions first.
- Monitor logs and performance in CloudWatch for regressions.

